Carl Andrew Ziegler

Exoplanet Astronomer | Associate Professor (August 2026) | Observatory Director

Department of Physics, Engineering, and Astronomy
Stephen F. Austin State University, Nacogdoches, TX 75962
(936) 244-8545
carl.ziegler@sfasu.edu
carlziegler.space
Google Scholar ▪ *Gravitas*

Professional Profile

Quantitative observational astrophysicist and university educator with more than a decade of experience designing measurements, extracting signals from noisy data, and translating evidence into validated survey products, scientific software, and learning experiences. Leads international high-resolution imaging programs for Kepler and TESS, builds Python-backed automated analysis pipelines, and investigates how stellar companions alter exoplanet detection and planetary-system architecture. Author or coauthor of more than 180 refereed publications with over 5,400 Google Scholar citations; mentor to more than 20 student researchers and Director of the SFA Observatory.

Research Impact

180+	5,400+	49	4,800+	20+
refereed publications	career citations	Google Scholar h-index	Kepler/TESS hosts imaged	student researchers mentored

Profiles: Google Scholar ▪ SFA Profile ▪ NASA ADS

Expertise

Research and Measurement: Experimental and survey design; instrument and evaluation development; uncertainty and error analysis; selection effects; hypothesis testing; Monte Carlo simulation; Bayesian modeling; image and time-series analysis.

Scientific Computing: Python, Astropy, emcee, JavaScript, HTML/CSS, C++, LabVIEW, and Mathematica; automated pipelines, statistical signal extraction, visualization, browser-based simulations, and robotic systems.

Learning and Product Practice: Authentic assessment, capability-growth measurement, curriculum prototyping, iterative feedback loops, project-based learning, research mentorship, and translation of research into deployed educational tools.

Academic Appointments

2020–Present **Assistant Professor of Astronomy; Associate Professor effective August 2026**, *Stephen F. Austin State University*, Nacogdoches, Texas

- **Director, SFA Observatory:** Lead research, instructional, and public-outreach operations; coordinate student training, observing programs, eclipse events, and partnerships with regional schools and astronomy organizations.
- **Teaching and assessment:** Build and validate examinations, laboratories, projects, and learning rubrics aligned with explicit objectives. Use student work, platform analytics, quantitative evaluations, and qualitative feedback to diagnose misconceptions and improve instruction.
- **Curriculum development:** Rebuilt Observational Astronomy as a semester-long authentic research experience and developed a graduate-style Astrophysics II course with original notes, problem sets, examinations, and worked solutions.
- **Research mentorship:** Mentored more than 20 undergraduates in Python, photometry, high-resolution imaging, and open-ended research, producing peer-reviewed coauthorships, conference presentations, REU placements, and technical career outcomes.

2018–Present **Principal Investigator and Quantitative Research Lead, SOAR TESS Survey**, *International collaboration across SFA, CTIO, UNC, and partner institutions*

- Lead large-scale empirical evaluation of TESS planet candidates using thousands of high-resolution observations; establish detection thresholds, quantify sensitivity and selection effects, remove false positives, and test population hypotheses under incomplete and noisy data.
- Engineer Python-backed workflows for calibration, image registration, point-spread-function subtraction, photometry, astrometry, quality control, and contrast-curve generation, converting raw observations into validated reusable data products.
- Demonstrated that close binary systems suppress detected planets by nearly a factor of seven and that uncorrected blending can bias inferred planet-occurrence rates by approximately a factor of two.
- Coordinate telescope operations, target prioritization, distributed collaborators, student contributors, data releases, and publication cycles.

2018–2020 **Dunlap Postdoctoral Fellow**, *Dunlap Institute for Astronomy and Astrophysics, University of Toronto*, Toronto, Ontario, Canada

- Founded the SOAR TESS Survey and developed automated target-selection and Python data-processing workflows for high-resolution follow-up of TESS Objects of Interest.
- Created the *One-Hit Wonders* program, moving long-period candidates from single-transit discovery through period prediction, robotic monitoring, signal recovery, and validation.
- Managed cross-institutional observing strategy, target queues, data handoffs, analysis responsibilities, and publication milestones across international partners.

Education

2018 **Ph.D. in Physics**, *University of North Carolina at Chapel Hill*, Chapel Hill, North Carolina
Dissertation: *Characterization of Exoplanets and Stellar Systems with New Robots*. Advisor: Nicholas Law.

2013 **M.S. in Physics**, *Southern Illinois University Carbondale*, Carbondale, Illinois
Thesis: *Adsorption of Neon on Open Carbon Nanohorn Aggregates*. Advisor: Aldo Migone.

2009 **B.A. in Physics and Mathematics**, *William Jewell College*, Liberty, Missouri
Undergraduate research in variable stars and globular clusters. Advisor: Maggie Sherer.

Selected Research Leadership and Contributions

SFA-era Productivity At the 2025 tenure review, authored or coauthored 136 refereed journal articles since joining SFA in 2020, more than 22% of the 617 refereed articles published university-wide during that period; these papers received more than 4,700 citations during the SFA appointment.

SOAR TESS Survey Founded and lead a systematic speckle-imaging survey on the 4.1-m SOAR Telescope. The survey has resolved companions to more than 1,000 targets, established population-level multiplicity constraints, and demonstrated the strong suppression of planets in close stellar binaries. Survey results.

Robo-AO-2 TESS Survey Co-designed a proposed high-cadence, simultaneous visible/near-infrared survey of approximately 4,000 TESS Objects of Interest. Robo-AO-2 can observe up to 200 targets per night; the program is expected to identify roughly 600 close binaries, correct diluted planet radii, and monitor orbital motion over a two-year baseline. Validated reduced products will be released through ExoFOP.

Robo-AO Kepler Survey Collaborators and I obtained high-resolution images of all 3,857 Kepler planet-candidate systems. These data identified blended stars and false positives, revised hundreds of planet radii, and supported the validation of more than 1,500 Kepler and K2 planets. Key survey paper.

Robo-SOAR Instrumentation Designed and built the testbed for a low-cost natural-guide-star adaptive-optics system for SOAR, pairing a reflective wavefront-sensor architecture with robotic control software to broaden access to high-resolution imaging.

Long-Period Planets — Developed robotic observing and precision-photometry strategies to recover later transits of long-period TESS candidates initially detected as single events, preserving high-value temperate planets for validation and atmospheric study.

Selected Peer-Reviewed Publications

Selected lead-author and major-contribution papers. For the complete and current record, see Google Scholar or the bibliography generated from `pubs.bib`.

2022 M. Salama, **C. Ziegler**, et al. *An Adaptive Optics Census of Companions to Northern Stars Within 25 pc with Robo-AO*. AJ, **163**, 5.

2021 **C. Ziegler**, et al. *Robo-AO and SOAR High-Resolution Surveys of Exoplanet-Hosting Stars*. *Frontiers in Astronomy and Space Sciences*, **8**, 625230.

2021 **C. Ziegler**, et al. *SOAR TESS Survey II: The Impact of Stellar Companions on Planetary Populations*. AJ, **162**, 192.

2020 **C. Ziegler**, et al. *SOAR TESS Survey I: Sculpting of TESS Planetary Systems by Stellar Companions*. AJ, **159**, 19.

2018 **C. Ziegler**, et al. *Measuring the Recoverability of Close Binaries in Gaia DR2 with the Robo-AO Kepler Survey*. AJ, **156**, 259.

2018 **C. Ziegler**, et al. *Robo-AO Kepler Survey V: The Effect of Physically Associated Stellar Companions on Planetary Systems*. AJ, **156**, 83.

2018 **C. Ziegler**, et al. *Robo-AO Kepler Planetary Candidate Survey IV: The Effect of Nearby Stars on 3,857 Planetary Candidate Systems*. AJ, **155**, 161.

2017 **C. Ziegler**, et al. *Robo-AO Kepler Planetary Candidate Survey III: Adaptive Optics Imaging of 1,629 Kepler Exoplanet Candidate Host Stars*. AJ, **153**, 66.

2016 C. Baranec, **C. Ziegler**, et al. *Robo-AO Kepler Planetary Candidate Survey II: Adaptive Optics Imaging of 969 Kepler Exoplanet Candidate Host Stars*. AJ, **152**, 18.

2015 **C. Ziegler**, et al. *Multiplicity of the Galactic Senior Citizens: A High-Resolution Search for Cool Subdwarf Companions*. ApJ, **804**, 30.

Selected Instrumentation Publications

2016 N. Law, **C. Ziegler**, and A. Tokovinin. *SRAO: The Southern Robotic Speckle plus Adaptive Optics System*. Proc. SPIE, **9907**, 99070K.

2016 **C. Ziegler**, et al. *SRAO: Optical Design and the Dual Knife-Edge Wavefront Sensor*. Proc. SPIE, **9909**, 99093Z.

2016 **C. Ziegler**, et al. *The Robo-AO KOI Survey: Laser Adaptive Optics Imaging of Every Kepler Exoplanet Candidate*. Proc. SPIE, **9909**, 99095U.

Funding and Awarded Observing Time

External Grant Proposals

2026 Co-Investigator, *Resolving the Multiplicity Bottleneck: Dual-Band Imaging of TESS Planetary Environments*, NASA TESS General Investigator Program. Approximately $300,000; submitted and decision pending. Responsible for TFOP coordination, automated target-list maintenance, and scientific interpretation. The proposal supports a graduate researcher and a public survey of approximately 4,000 TOIs using Robo-AO-2.

2024 Co-Principal Investigator, *Advancing Autonomous Adaptive Optics and Adaptive Secondary Mirror Technologies*, National Science Foundation. $300,000. Submitted.

2023–2024 Co-Investigator, *A Uniform and Systematic High-Acuity Survey of TESS Objects of Interest*, NASA TESS General Investigator Program. Approximately $250,000. Submitted; rated "Very Good."

Awarded Telescope Time as Principal Investigator

2022A SOAR Telescope, 3 nights. *SOAR TESS Survey of Exoplanet Candidate Hosts*.

2021A SOAR Telescope, 4 nights. *Characterization of TESS Planets in Multiple-Star Systems*.

2020B SOAR Telescope, 3 nights. *SOAR TESS Survey: Characterization of TESS Planets*.

Selected Scholarly Software and Educational Innovation

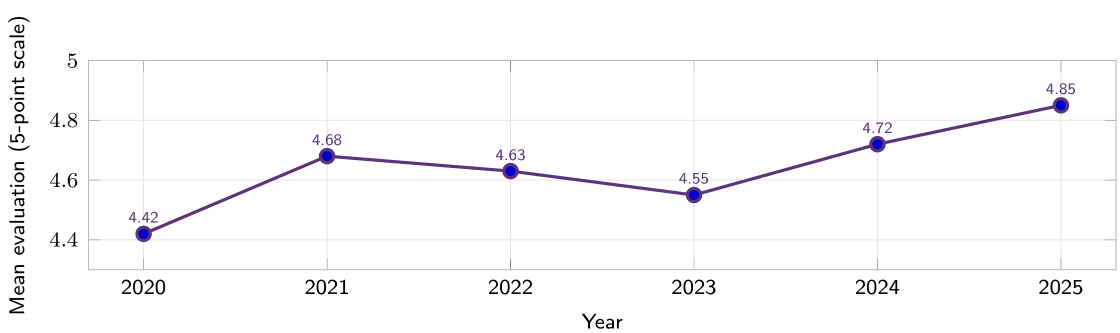
2025–Present **Creator and Lead Developer, Gravitas**, *Open-source browser-based astronomy simulation*

- Designed, coded, and deployed an interactive sandbox for students and the public to construct orbital systems, vary physical parameters, and test predictions through real-time visualization; co-developed the project with an undergraduate researcher as a lasting departmental demonstration and learning resource.
- Implemented modules for orbital and black-hole dynamics, energy conservation, compact-object mergers, spacetime structure, exoplanet transit light curves, and gravitational-wave sonification.
- Integrated Gravitas into SFA introductory astronomy as a guided-inquiry tool in which learners predict, test, revise, and explain rather than passively view animations.
- Public application: gravitas-sim.online; source code and documentation: GitHub repository.

Teaching and Mentorship

Documented Teaching Effectiveness

Evaluations 2025 annual average: **4.85/5**. Across 10 semesters, overall evaluations regularly exceeded 4.5/5; items addressing timely grading, civility, and accessibility averaged above 4.7/5.



Peer and Administrative Review Formal CTL peer observation and annual administrative reviews document sustained growth from an "enthusiastic teacher" to an "exemplary teacher," with the most recent review highlighting thoughtful pedagogical innovation.

Courses Taught and Developed at Stephen F. Austin State University

Astronomy Modern and Classical Astronomy (ASTR 1303) ▪ Astronomy Laboratory (ASTR 1103) ▪ Observational Astronomy (ASTR 3305) ▪ Advanced Observational Astronomy Laboratory (ASTR 3005) ▪ Astrophysics (ASTR 3335)

Physics Mathematical Methods in Physics (PHYS 3347) ▪ Senior/Physics Practicum (PHYS 4170)

Directed Study Independent Study in Physics and Astronomy (PHYS 4175/4176) ▪ M.S. Thesis Research (PHYS 5389/5390)

Selected Course Design and Pedagogical Innovation

- **Introductory Astronomy:** Redesigned online and face-to-face instruction around a clear orientation, short concept videos, low-stakes checks, targeted mathematics practice, growth-mindset messaging, and proactive individual outreach; incorporated Gravitas and guided inquiry to test orbital, energy, black-hole, and transit models.

- **Observational Astronomy:** Rebuilt the course as a semester-long authentic capability assessment in which students formulate questions, write observing proposals, collect telescope data, calibrate and analyze observations, evaluate uncertainty, and communicate results professionally.
- **Astrophysics II:** Developed a complete upper-level course with original lecture notes, problem sets, examinations, and worked solutions connecting stellar structure and evolution, galaxies, cosmology, and compact objects to quantitative modeling and current research.
- **Professional formation:** Coach students through CVs, personal statements, graduate-school preparation, scientific presentations, interviews, and project management through Senior Practicum and research mentoring.
- Earned SFA Center for Teaching and Learning Certified Online Instructor status (2020) and completed four Association of College and University Educators courses (2024), including certifications in *Promoting Active Learning Online* and *Inspiring Inquiry and Lifelong Learning in Your Online Course*; applied these practices through shorter modules, formative feedback, collaborative work, and the SFA Student Experience Project.
- Created two 12-lecture radiation and electromagnetism courses for the SFA STEM Academy, including hands-on activities, worksheets, guided notes, solutions, and turnkey instructor resources.
- Developed and shared comprehensive introductory-astronomy notes, slide decks, assignments, and examinations with new lecturers, and coordinated observatory night laboratories and cross-disciplinary astrophotography activities for colleagues.

Research Mentorship

- Mentored more than 20 undergraduate students and multiple graduate students in exoplanet photometry, automated transit recovery, Python data pipelines, stellar multiplicity, high-resolution imaging, and scientific software.
- Guided students to peer-reviewed coauthorship, presentations at American Astronomical Society meetings, REU placements, graduate-school preparation, and technical careers, including a NASA flight-controller role.
- Supervised projects including *Automated Recovery of TESS Single-Transit Planets*, *Photometric Follow-up of Transiting Planets*, and *Analysis of High-Resolution Images of Exoplanet Host Stars*.

Selected Invited and Contributed Presentations

2023 *SOAR TESS Speckle Survey of Exoplanet Candidates*. TSAAPT/TSAPS Meeting, Texas A&M University–Commerce.

2020 *SOAR TESS Survey: The Sculpting of Planetary Systems by Stellar Companions*. 235th Meeting of the American Astronomical Society, Honolulu, Hawaii.

2019 *One-Hit Wonders: Hunting the Longest-Period TESS Planets*. TESS Science Conference I, Cambridge, Massachusetts.

2019 *One-Hit Wonders: Hunting the Longest-Period TESS Planets*. Canadian Astronomical Society Annual Meeting, Montreal, Quebec.

2019 *Death Stars? How Tight Binaries Impact TESS Planets with SOAR Speckle Imaging*. 233rd Meeting of the American Astronomical Society, Seattle, Washington.

2018 *Robo-AO KOI Survey: Laser-Guide-Star Adaptive-Optics Imaging of Every Kepler Planetary-Candidate Host Star*. 231st Meeting of the American Astronomical Society, National Harbor, Maryland.

2017 *High-Resolution Imaging of Four Thousand Kepler Planetary-Candidate Host Stars*. Know Thy Star, Know Thy Planet Conference, Pasadena, California.

2016 *The Robo-AO KOI Survey and the Development of a Southern Robotic Adaptive-Optics System*. Invited seminar, Institute for Astronomy, Hilo, Hawaii.

Service and Leadership

University Service

2023–Present Member, College of Sciences and Mathematics College Council; participated in curriculum oversight and review of more than 200 curriculum changes across the college.

2023–Present College of Sciences and Mathematics representative, University Grievance Board.

2024–Present Member, Physics Degree Revision and Modification Committee; contributed to plans for Texas Physics Consortium participation and early development of a Medical Physics track.

2022 Chair, Expectations and Values Committee; led development of a departmental framework for shared professional expectations and respectful communication.

2021–Present Member, Recruitment, Retention, and Outreach Committee; established observatory alumni nights and used the facility as a recruitment and community-building resource.

Professional and Community Service

NASA Review Panelist on three NASA grant-review committees evaluating proposals for major missions and competitive research programs.

Journal Review Referee for MNRAS, AJ, ApJ, PASP, and A&A.

Observatory Outreach Personally organized more than 20 public observing nights since 2021, reaching over 2,000 visitors; free tickets are frequently claimed within hours. Led a 2023 annular-eclipse event for more than 200 attendees and supported the 2024 total-eclipse student trip and public programming. SFA Observatory.

Regional STEM Engagement Developed two 12-lecture STEM Academy curricula; host Scout troops, summer camps, regional schools, LeTourneau University classes, and other community groups; coordinate programming with the Astronomical Society of East Texas.

Public Communication Created an eclipse-safety and science video promoted through SFA channels and regularly speak for schools, amateur astronomy organizations, and community groups.

Technical Expertise

Programming and Web Python, JavaScript, HTML/CSS, Astropy, emcee, C++, LabVIEW, and Mathematica; scientific visualization, automated pipelines, browser-based simulations, and instrument-control software.

Research Methods Experimental and survey design, Bayesian modeling, Monte Carlo simulation, uncertainty analysis, hypothesis testing, selection-function analysis, mixed-methods evaluation, and reproducible workflows.

Instrumentation Optical and opto-mechanical design with Zemax and SolidWorks; robotic telescope systems; adaptive optics; speckle interferometry; Shack–Hartmann, pyramid, and reflective knife-edge wavefront sensing.

Observing and Analysis SOAR Telescope operations; TheSkyX and MaxIm DL; fast-frame FITS-cube reduction, time-series photometry, image registration, point-spread-function subtraction, companion detection, astrometry, contrast-curve generation, catalog cross-matching, ExoFOP data delivery, and population-level statistical inference.

Professional References

Suresh Sivanandam — Director, Dunlap Institute, U. Toronto
sivanandam@dunlap.utoronto.edu ▪ 416-978-6550

Christoph Baranec — Institute for Astronomy, U. Hawai'i at Mānoa
baranec@hawaii.edu ▪ 808-932-2318

Nicholas Law — Physics & Astronomy, UNC–Chapel Hill
nlaw@unc.edu ▪ 919-962-3019

Adam Kraus — Astronomy, The University of Texas at Austin
alk@astro.as.utexas.edu ▪ 617-956-7740